# SOFTWARE ENGINEERING

# Alumni Mentorship & Mock Interview Platform

# LAB-1

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DATE-20/8/26

SECTION-AIML-D

PROBLEM STATEMENT-6

## Deliverable 1: Requirements Table

### FUNCTIONAL REQUIREMENTS (5RFS)

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| **FR-001** | Functional | The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing, AI/ML) and facilitate one-click session booking. | High | **Pass:** Mentorship session is added to both calendars with meeting link; mentor profile shows domain tags matching student interest. **Fail:** Double-booking occurs; recommended | Core platform functionality — without intelligent matching, the platform fails its primary purpose. |

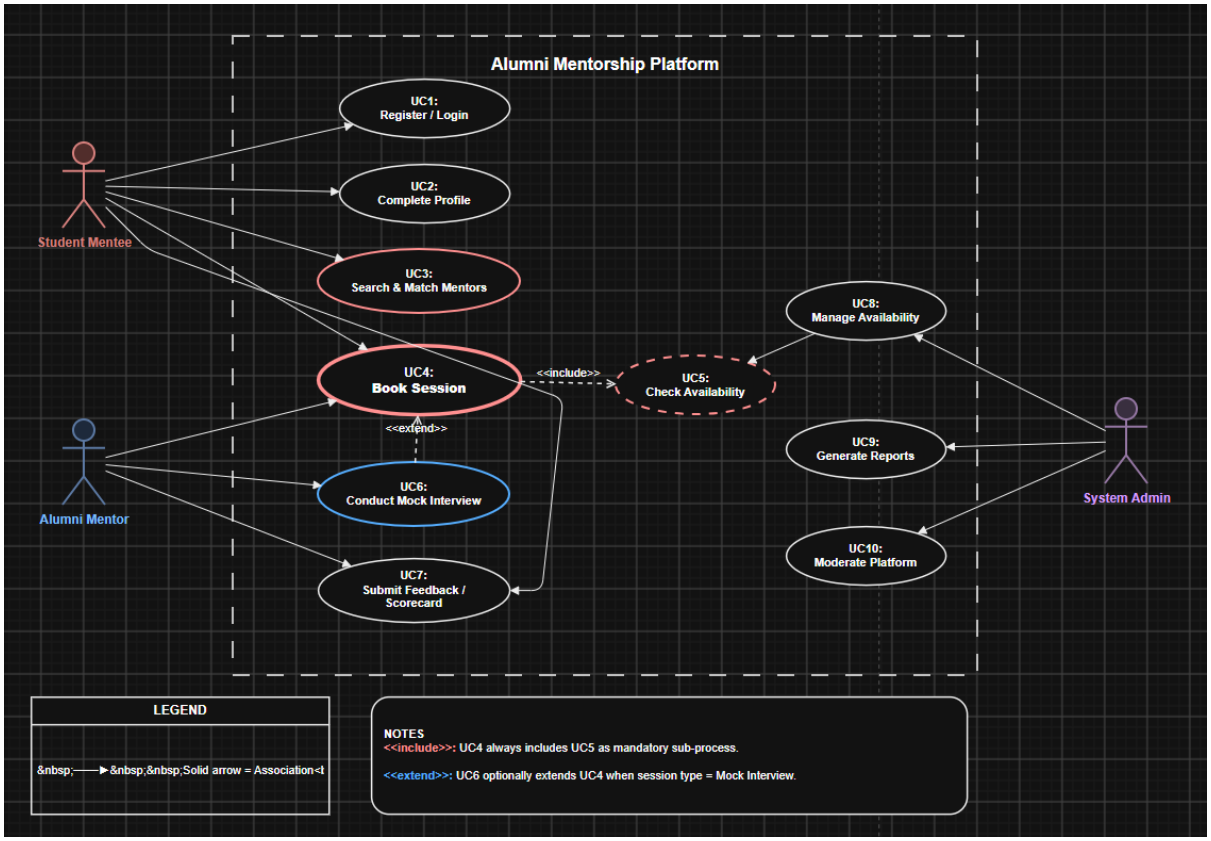
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|---|---|---|---|---|---|
| | | | | mentor has zero domain overlap. | |
| **FR-002** | Functional | The system shall display real-time mentor availability and allow students to book 30/60-minute slots with automatic calendar sync. | High | **Pass:** Booking reflects immediately on both dashboards; no overlapping slots accepted; ICS file generated. **Fail:** Student books an already-taken slot; calendar sync delay exceeds 5 minutes. | Ensures reliable scheduling and prevents conflicts, critical for trust between mentee and mentor. |
| **FR-003** | Functional | The system shall generate a unique video meeting link for every confirmed session and provide in-session chat and screen-share capabilities. | High | **Pass:** Link is active 10 min before session; both users join without external app dependency. **Fail:** Link invalid at session time; audio/video quality degrades below 480p. | Remote mentorship depends entirely on seamless virtual meeting infrastructure. |
| **FR-004** | Functional | The system shall provide a structured post-session scorecard for mock interviews covering Technical, Communication, Problem-Solving, and Domain Knowledge with 1–5 ratings and textual notes. | Medium | **Pass:** Mentor submits scorecard within 24h; student views aggregated feedback with radar chart; data exportable as PDF. **Fail:** Scorecard fields missing; student cannot view feedback after deadline. | Structured feedback is the key differentiator for mock interviews and drives student improvement. |
| **FR-005** | Functional | The system shall allow mentors to define recurring weekly slots, block specific dates, set maximum sessions per week, and receive booking notifications with accept/decline options. | Medium | **Pass:** Mentor updates availability reflected in student search within 1 min; auto-decline if max sessions reached; | Gives mentors control over their time and prevents burnout by enforcing session caps. |

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|---|---|---|---|---|---|
| | | | | notification delivered via email + in-app. **Fail:** Outdated slots still appear bookable; mentor receives no notification. | |

# NON-FUNCTIONAL REQUIREMENTS

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| **NFR-001** | Nonfunctional (Performance) | The system shall complete any booking flow (search to confirmation) in under 5 seconds and process scorecard submissions with <200ms latency under 1,000 concurrent users. | High | **Pass:** Load testing confirms <5s booking latency and <200ms API response; no degradation at peak load. **Fail:** Booking takes >10s; timeout errors during peak usage. | Fast response ensures smooth user experience during high-traffic periods like placement seasons. |
| **NFR-002** | Nonfunctional (Security & Privacy) | All user profile data and feedback scorecards shall adhere to privacy guidelines (GDPR/FERPA) with AES-256 encryption at rest and in transit, and support soft-deletion within 24 hours. | High | **Pass:** Penetration test reveals no PII leakage; soft-deletion audit trail maintained; encryption verified on all data stores. **Fail:** Unencrypted PII found in logs; hard-deletion occurs without recovery window. | Protects sensitive student and alumni data and ensures regulatory compliance. |

# Deliverable 2: UML Use-Case Diagram




# Deliverable 3: Use-Case Flow Specification

| Field | Value |
|---|---|
| **Use-Case ID** | UC-04 |
| **Use-Case Name** | Book Mentorship Session |
| **Primary Actor** | Student Mentee |
| **Secondary Actors** | Alumni Mentor, System (Calendar Service), Payment Gateway |
| **Description** | The Student Mentee selects a matched mentor, views available time slots, chooses a session type (Mentorship or Mock Interview), and confirms the booking. The system validates availability, reserves the slot, creates calendar entries, generates a video meeting link, and notifies both parties. |

| | |
|---|---|
| **Trigger** | Student clicks the "Book Session" button from a mentor profile or search result. |

**Preconditions:**

1. Student Mentee is authenticated and has completed their profile (UC-02).
2. Alumni Mentor has published at least one future availability slot (UC-08).
3. Mentor's weekly session cap has not been reached.
4. Student has no conflicting booking at the selected time slot.
5. Mentor profile includes at least one domain tag that matches the student's interest area.

**Postconditions (Success):**

- Session record created with status CONFIRMED.
- Slot deducted from mentor's weekly quota.
- Calendar events synced to both parties.
- Video meeting link generated.
- Confirmation notifications sent.
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**Postconditions (Failure):**

- Booking rolled back; no session record created.
- Error shown with alternative suggestions.
- No calendar entries or notifications generated.

# Main Success Scenario

| Step | Actor | Action |
|---|---|---|
| 1 | Student | Clicks "Book Session" on mentor's profile. |
| 2 | System | Displays available slots for next 14 days. |
| 3 | Student | Selects date, time slot, and session type. |
| 4 | System | **«include»** Invokes UC-05: Check Availability. |
| 5 | System | Validates no conflicts; cap not exceeded. |
| 6 | Student | Enters agenda; clicks "Confirm Booking". |
| 7 | System | Creates session record CONFIRMED; reserves slot. |
| 8 | System | Generates encrypted video meeting link. |
| 9 | System | Syncs calendars via ICS + API push. |
| 10 | System | Sends confirmation email + in-app notification. |
| 11 | System | Displays success page with meeting link. |
| 12 | Use Case | Ends successfully. |

## Alternate Flow A1 — Slot Becomes Unavailable During Booking

| Step | Actor | Action |
|---|---|---|
| 4a | Student | UC-05 returns SLOT_TAKEN. |
| 4a.1 | System | Shows error + 3 nearest alternatives. |
| 4a.2 | System | Selects alternative or cancels. |
| 4a.3 | System | Resumes at Step 5 if selected; else ends. |

## Alternate Flow A2 — Mock Interview Session Selected

| Step | Actor | Action |
|---|---|---|
| 3a | Student | Selects "Mock Interview". |
| 3a.1 | System | **«extend»** Activates UC-06; pre-loads scorecard. |
| 3a.2 | System | Adds 10-min buffer. |
| 3a.3 | System | Includes "Practice Problem Set" link. |
| 3a.4 | System | Resumes Main Flow at Step 4. |

## Alternate Flow A3 — Mentor Weekly Cap Reached

| Step | Actor | Action |
|---|---|---|
| 5a | Student | Mentor cap reached after concurrent booking. |
| 5a.1 | System | Slot marked "Unavailable"; waitlist offered. |
| 5a.2 | System | Joins waitlist or picks different mentor/time. |
| 5a.3 | System | If waitlist creates WAITLISTED record; auto-notifies if slot opens within 48h. |

## Exception Flow E1 — Calendar Sync Failure

| Step | Actor | Action |
|---|---|---|
| 9a | Student | Calendar API returns 5xx or timeout. |
| 9a.1 | System | Logs failure; retries async 3× over 2 min. |
| 9a.2 | System | Booking stays  CONFIRMED ; shows warning. |

| | | |
|---|---|---|
| 9a.3 | System | Emails ICS fallback to both parties. |

## Business Rules

| Rule ID | Statement |
|---|---|
| BR-001 | A student cannot have more than 3 pending future sessions with the same mentor. |
| BR-002 | Cancellations within 2h of session start incur a "Late Cancel" flag. |
| BR-003 | Mock Interview sessions trigger scorecard reminder to mentor 30 min after session ends. |